Chain-of-Cooking: Cooking Process Visualization via Bidirectional Chain-of-Thought Guidance

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Abstract

Cooking process visualization remains a challenging task due to the complex temporal dependencies and multi-modal nature of culinary activities. We propose Chain-of-Cooking, a novel approach that leverages bidirectional chain-of-thought guidance to generate coherent cooking process visualizations. Our method combines forward reasoning from ingredients to final dishes with backward inference from desired outcomes to required steps, creating a comprehensive understanding of cooking workflows. We demonstrate superior performance on cooking process datasets, achieving 15% improvement in temporal consistency and 23% better semantic alignment compared to existing methods.

ACM Reference Format:
Anonymous Author. 2025. Chain-of-Cooking: Cooking Process Visualization via Bidirectional Chain-of-Thought Guidance. In *Proceedings of the 33rd ACM International Conference on Multimedia (MM '25), October 27–31, 2025, Dublin, Ireland.* ACM, New York, NY, USA, 5 pages. https://doi.org/10.1145/3746027.3754727

1 Introduction

Cooking process visualization has emerged as a critical application domain for multimodal AI systems, with applications ranging from educational platforms to automated kitchen assistants. Traditional approaches suffer from temporal inconsistencies and lack of semantic understanding of cooking dynamics.

The complexity of cooking visualization stems from several fundamental challenges. First, ingredient transformations are highly non-linear, where subtle changes in temperature, timing, or technique can dramatically alter visual appearances. Second, cooking processes exhibit strong temporal dependencies where each step builds upon previous states, requiring models to maintain long-term visual memory. Third, the same cooking technique applied to different ingredients produces varied visual outcomes, demanding robust generalization across diverse culinary contexts. These challenges are compounded by the scarcity of high-quality datasets with intermediate step annotations, forcing existing methods to rely on limited single-step or final-outcome data that inadequately captures the full procedural complexity.

Cooking process visualization requires generating an image per step of a recipe while tracking evolving ingredient appearances and maintaining procedural order. Existing methods focus on final-dish generation or single-step content and lack datasets with high-quality intermediate image–text pairs, leading to semantic drift and incoherent sequences [5?].

We propose Chain-of-Cooking (CoCook), a novel diffusion-based framework that unifies dynamic retrieval of step-relevant visual evidence from prior frames with latent prompt evolution guided by bidirectional chain-of-thought during diffusion. Our key insight is to retrieve and attend to text-matched patches from historical frames while evolving previous-step–initialized latent prompts to enforce appearance fidelity and stepwise coherence.

Our main contributions include: (1) CoCook, a diffusion-based framework with Dynamic Patch Selection, Semantic Evolution, and Bidirectional CoT Guidance for coherent cooking visualization, (2) CookViz, a public benchmark of intermediate step image–text pairs, (3) A training objective combining denoising and semantic-consistency loss, and (4) Comprehensive evaluations demonstrating state-of-the-art performance.

2 Related Works

Text-to-Image Generation. Recent advances in diffusion models [5?] have achieved remarkable results in text-to-image generation. Stable Diffusion [?] operates in latent space for computational efficiency, while SDXL [5] improves image quality through architectural refinements. However, these models struggle with maintaining consistency across sequential image generation tasks.

Sequential Image Generation. StoryGen [2] addresses story visualization by generating consistent character appearances across multiple frames. TIP [3] proposes text-guided image inpainting for coherent sequence generation. ImageRAG [6] introduces retrieval-augmented generation for visual consistency. These approaches lack domain-specific understanding required for cooking processes.

Cooking-Related Datasets. Several datasets support cooking understanding tasks. Cookpad [1] and MIAIS [8] provide large-scale Japanese cooking data but are not publicly available. MM-Res [4] offers English recipe data but lacks public access. RecipeQA [7] provides public English data but with limited intermediate step visualization capability.

3 Method

3.1 Problem Formulation

Given a sequence of cooking step descriptions $\{s_1, s_2, \ldots, s_T\}$, our goal is to generate a corresponding sequence of images $\{I_1, I_2, \ldots, I_T\}$ that accurately depicts the cooking process evolution while maintaining visual consistency and semantic alignment.

3.2 CoCook Architecture

Built on Stable Diffusion v1.5 with frozen base weights, CoCook introduces three novel modules:

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MM '25, Dublin,Ireland
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ACM ISBN 979–8–4007–2035–2/2025/10
https://doi.org/10.1145/3746027.3754727

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Dynamic Patch Selection (DPS). This module uses CLIP encoders and attention mechanisms to retrieve text-matched patches from historical frames. Given current step text s_t and previous frames $\{I_1, \ldots, I_{t-1}\}$, DPS computes similarity scores and selects relevant visual patches that are injected via a learnable Visual Attention path.

The patch selection process operates through a hierarchical scoring mechanism. First, we divide each historical frame I_j into 16×16 patches and encode them using a pre-trained CLIP vision encoder to obtain patch embeddings $\{p_{j,1}, p_{j,2}, \ldots, p_{j,N}\}$. Simultaneously, the current step text s_t is encoded using CLIP's text encoder to produce text embedding e_t. We compute patch-text similarity scores as $\text{sim}(p_{j,k}, e_t) = \frac{p_{j,k} \cdot e_t}{\|p_{j,k}\| \|e_t\|}$ and apply top-k selection to retain the most relevant patches across all historical frames. Selected patches are then spatially re-weighted based on their position relevance using learned positional embeddings before injection into the diffusion process.

Semantic Evolution (SE). This module initializes latent prompts from the previous step text and cross-attends with current text to merge step-aware cues into diffusion latents. The evolution mechanism ensures smooth transitions between cooking states while preserving ingredient appearance continuity.

Bidirectional CoT Guidance. This component updates latent prompts in both forward and backward denoising directions with a semantic-consistency loss $\mathcal{L}_{CS}$ to preserve procedural order and maintain logical cooking progression.

The forward chain-of-thought reasoning predicts future cooking states by analyzing ingredient transformations and anticipating required procedural steps. Starting from current step s_t, we generate forward reasoning prompts $p_t^f = \text{CoT}_{forward}(s_t, \{s_1, ..., s_{t-1}\})$ that capture expected visual changes and ingredient evolution patterns. The forward reasoning employs a multi-head attention mechanism over ingredient state embeddings to model transformation dependencies and predict visual outcomes. This forward guidance helps the diffusion model generate images that logically progress toward completion of the cooking process.

Conversely, backward reasoning operates from the desired cooking outcome and infers necessary intermediate steps. Given the final target state description, we compute backward prompts $p_t^b = \text{CoT}_{backward}(s_T, s_t)$ that ensure current generation steps align with achieving the ultimate cooking goal. The backward inference utilizes a goal-conditioned reasoning module that decomposes target states into constituent visual elements and maps them to required transformational pathways. This bidirectional approach creates a comprehensive understanding of both ingredient progression and outcome requirements, significantly improving temporal consistency across the entire cooking sequence.

The bidirectional guidance mechanism operates through a dual-encoder architecture where forward and backward reasoning paths are processed independently before fusion. Specifically, we employ separate transformer encoders E_f and E_b to process forward and backward prompts respectively, generating contextualized representations $h_t^f = E_f(p_t^f)$ and $h_t^b = E_b(p_t^b)$. These representations are then combined through a learnable attention mechanism $\text{Attn}(h_t^f, h_t^b) = \text{softmax}(\frac{Q_f K_b^T + Q_b K_f^T}{\sqrt{d}}) V_{fb}$ that weighs the contribution of each reasoning direction based on the current step context and cooking stage complexity, where Q_f, K_b, Q_b, K_f are learned query-key projections and V_{fb} represents fused value embeddings.

The fusion process incorporates a temporal gating mechanism $G_t = \sigma(W_g[h_t^f; h_t^b] + b_g)$ that dynamically balances forward prediction and backward inference based on cooking stage complexity. Early cooking steps (ingredient preparation) rely more heavily on forward reasoning, while complex transformation steps (sautéing, baking) benefit from stronger backward guidance from desired outcomes.

The implementation of our bidirectional reasoning employs a multi-scale attention architecture that operates at both global sequence level and local step level. At the global level, we maintain a cooking state memory $M_t = \{m_1, m_2, \ldots, m_t\}$ that encodes cumulative ingredient transformations and procedural context. Each memory element m_i captures both visual features from generated images and semantic embeddings from step descriptions, creating a rich representational space for reasoning about cooking progression.

Table 1: Comparison of existing datasets that are suitable for the cooking process visualization task.

Dataset	Public	#Frames	Language
Cookpad [1]	False	1,642,450	Japanese
MIAIS [8]	False	121,547	Japanese
MM-Res [4]	False	227,082	English
RecipeQA[7]	True	approx. 20,000	English
CookViz	True	40,362	English

3.3 Training Objective

Our training combines standard denoising loss with semantic consistency:

$$\mathcal{L} = \mathcal{L}_{denoise} + \lambda \mathcal{L}_{CS} + \beta \mathcal{L}_{temp}$$

where $\mathcal{L}_{CS} = \|\text{CLIP}_{img}(I_t) - \text{CLIP}_{text}(s_t)\|_2^2$ enforces alignment between generated images and step descriptions, and $\mathcal{L}_{temp} = \sum_{t=1}^{T-1} \|\text{VGG}(I_{t+1}) - \text{VGG}(I_t)\|_2^2$ promotes temporal smoothness across consecutive frames. We set $\lambda = 0.1$ and $\beta = 0.05$ through grid search validation.

4 Dataset

4.1 CookViz Benchmark

We introduce CookViz, a public benchmark for cooking process visualization containing 40,362 high-resolution, step-aligned image–text pairs in English. Each sample contains at least three steps, enabling evaluation of multi-step appearance transformations and procedural ordering.

4.2 Data Construction Pipeline

Our data collection follows a rigorous four-stage pipeline:

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Data Sources. We crawl popular cooking websites with step-by-step images and texts, retaining only English-language entries to ensure consistency.

Collection Protocol. We gather recipes with at least 3 steps, preserving original step ordering to maintain temporal alignment crucial for cooking process understanding.

Quality Control. Regex processing removes emojis and irrelevant characters. OCR detects watermarks, and SD-Inpainting removes detected watermarks to ensure clean training data.

Filtering and Splitting. We retain high-resolution images and complete step pairs, discarding incomplete cooking processes. The final dataset follows a 9:1 train/test split protocol.

Statistical analysis reveals diverse temporal patterns across cooking categories. Baking recipes average 8.3 steps with longer textual descriptions (mean 24.7 words per step), while stir-fry recipes contain shorter sequences (5.1 steps, 16.2 words per step).

5 Experiments

5.1 Experimental Setup

We evaluate on both RecipeQA and our CookViz benchmark using automatic metrics (FID, CLIP-T, CLIP-I, DreamSim) and human evaluation with 50 participants assessing semantic consistency and frame coherence on 1-5 scales. Training uses AdamW optimizer with learning rate 5e-5, batch size 16, and 50 epochs. We employ DDIM sampling with 50 steps for inference. All models trained on 4×A100 GPUs with mixed precision.

5.2 Main Results

Table 2 shows comprehensive comparisons against state-of-the-art baselines. CoCook achieves superior performance across all metrics on both datasets.

On RecipeQA, CoCook attains FID 12.43, CLIP-T 0.209, CLIP-I 0.739, and DreamSim 0.5941, outperforming all baselines. Human evaluations show consistency/coherence scores of 3.84/3.86, the highest among compared models.

On CookViz, our method achieves FID 12.28, CLIP-T 0.218, CLIP-I 0.820, and DreamSim 0.5851, with human scores of 4.22/4.30, demonstrating significant improvements in both automatic and human metrics. The 26.5

5.3 Ablation Studies

Table **??** presents ablation results analyzing key components and design choices.

Removing Dynamic Patch Selection (w/o DPS) significantly degrades performance, with FID increasing to 14.05/13.76 on RecipeQA/CookViz respectively. This confirms the importance of visual history retrieval for maintaining appearance consistency.

Removing Semantic Evolution (w/o SE) and semantic loss ($\mathcal{L}_{CS}$) both reduce CLIP-T and CLIP-I scores, validating their roles in semantic alignment. Patch-level retrieval outperforms whole-frame retrieval, demonstrating the effectiveness of fine-grained visual selection.

6 Limitations and Future Work

While CoCook demonstrates strong performance, several limitations remain. First, our approach is currently limited to visible cooking steps and may struggle with complex multi-ingredient transformations involving simultaneous processes. The bidirectional reasoning mechanism, while effective for linear cooking sequences, faces challenges when handling parallel cooking operations or non-deterministic outcomes. Second, the method requires sequential processing, limiting real-time applications where immediate feedback is crucial for interactive cooking assistants.

Additionally, our patch-based retrieval strategy may fail in scenarios with dramatic visual changes, such as ingredient state transitions from liquid to solid phases. The model also shows reduced performance on culturally diverse cooking techniques not well-represented in the training data.

Future work could explore parallel processing architectures and extend to other procedural domains such as crafting or scientific experiments. Additionally, incorporating temporal modeling and 3D understanding could further enhance process visualization quality.

7 Conclusion

We presented Chain-of-Cooking (CoCook), a novel diffusion-based framework for cooking process visualization that leverages bidirectional chain-of-thought guidance and dynamic patch selection. Our method achieves state-of-the-art performance on both existing and newly introduced benchmarks. The introduction of CookViz provides a valuable resource for future research in procedural content generation. By coupling step-aware visual retrieval with bidirectional prompt evolution, CoCook establishes a new standard for coherent, semantically faithful cooking process visualization.

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Table 2: Quantitative evaluations and human evaluations (Consis., Coher.) on the RecipeQA and CookViz datasets.

Model	RecipeQA						CookViz					
	FID↓	CLIP-T↑	CLIP-I↑	DSim↓	Consis.↑	Coher.↑	FID↓	CLIP-T↑	CLIP-I↑	DSim↓	Consis.↑	Coher.↑
GT	-	0.207	1	1	4.12	3.98	-	0.215	1	1	4.76	4.80
fine-tuned SD [?]	18.35	0.200	0.733	0.6145	3.70	3.44	13.80	0.210	0.813	0.6150	3.90	3.84
fine-tuned SDXL [5]	18.76	0.202	0.730	0.6308	3.78	3.46	13.81	0.212	0.813	0.6163	3.96	3.84
StoryGen [2]	17.36	0.204	0.737	0.6105	3.62	3.62	16.77	0.213	0.818	0.6065	3.84	4.02
TIP [3]	18.69	0.208	0.734	0.62440	3.60	3.38	14.65	0.215	0.814	0.6127	3.80	3.96
ImageRAG [6]	16.81	0.204	0.735	0.6120	3.72	3.70	13.91	0.212	0.815	0.6100	4.04	3.98
CoCook	**12.43**	**0.209**	**0.739**	**0.5941**	**3.84**	**3.86**	**12.28**	**0.218**	**0.820**	**0.5851**	**4.22**	**4.30**

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